

Data Lake Design for Kenya Mobile Money Transactions

GROUP: 5

MEMBERS:

1. Khaemba Imelda Nasimiyu - ST62/55266/2025
2. Grace Mwai - ST62/56407/2025
3. Murangiri Julius Elijah - ST62/56956/2025
4. Frieze Wandabwa - ST62/55175/2025
5. Mbarak Abdallah Kumbo - ST62/55566/2025
6. Seth Makori - ST62/55318/2025
7. Victor Chiuri Gateru - ST62/80979/2024
8. Paul Maweu Manthi - ST62/56199/2025
9. Erick Ngari - ST62/56218/2025
10. Framul Muriithi - ST62/55824/2025
11. Kariuki James Kariuki - ST62/55039/2025
12. Marrion Kiprop - ST62/80971/2024

1. Introduction

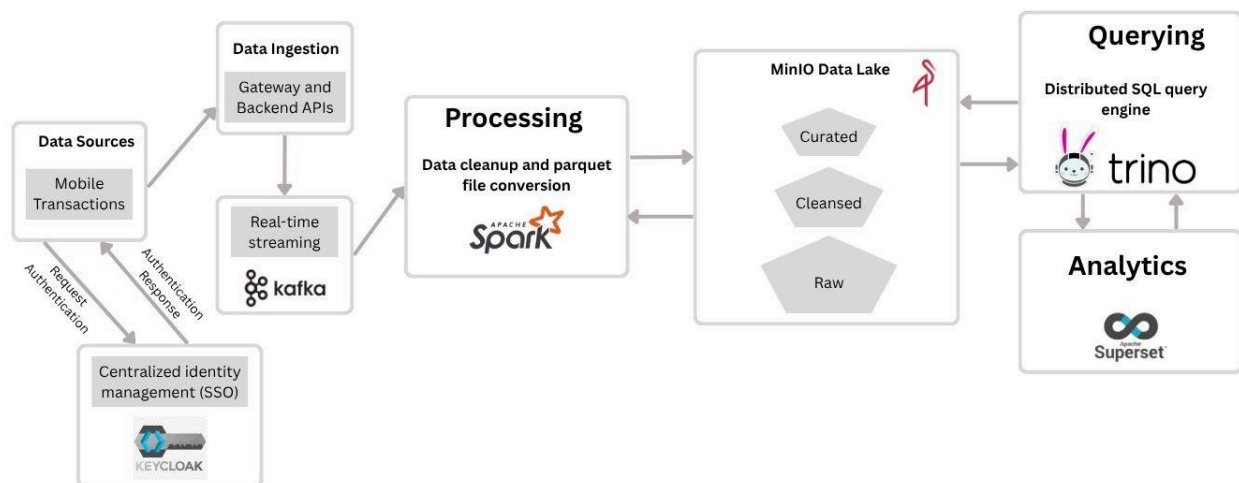
This project designs a three-zone data lake architecture to ingest, process, and analyze mobile money transactions in Kenya (e.g., transfers, withdrawals, paybill payments) to enable real-time fraud detection, user behavior analytics, and financial risk insights.

2. Data Lake Architecture

The architecture follows the medallion architecture (Raw → Cleansed → Curated), a proven pattern for progressive data quality. The following is the list of tools used:

- Apache Kafka (for streaming ingestion) - It ensures fault tolerance, message durability, batch flushing (every 10 seconds or after batching 500 events), and retry mechanisms
- Apache Spark (for batch data processing during data cleanup and data conversion to parquet file format)
- MinIO (data lake storage)
- Trino (analytics)
- Apache Superset (dashboards)

Data Lake Architecture



2.1 Zone Descriptions

Raw / Ingestion Zone:- Stores raw JSONL transaction events from Kafka to provide data traceability and replay capability.

Cleansed / Staging Zone:- Stores cleansed and validated data to ensure data quality and consistency

Curated / Analytics Zone:- Stores data in Parquet format, ensuring the data is well optimized for fast querying and Business Intelligence tools.

3. Deployment - Justification for On-Prem Data Lake:

1. Data Sovereignty & Regulatory Compliance - Mobile money data contains sensitive financial and personal information (PII), and keeping data on-prem ensures compliance with the Kenya Data Protection Act 2019
2. Enhanced Data Security - On-prem systems allow strict network-level controls (firewalls, private networks) and isolation from public cloud exposure
3. Cost Efficiency (Long-Term) - Avoids recurring cloud costs such as storage fees, data egress charges, compute billing, etc.
4. Low Latency & Real-Time Processing - On-prem deployment ensures data can be hosted in the same users' geographical location, reducing the number of network hops to cloud regions and thus enabling faster ingestion and processing.
5. Full Infrastructure Control- It ensures we have full control over hardware, storage configurations, security policies, etc making it possible to do custom tuning and better performance optimization
6. No Internet Dependency - Using on-prem ensures we can deploy in remote areas, low-bandwidth regions, and unreliable connectivity environments.

4. Analytics and Fraud Detection

The data lake supports both real-time and batch analytics to derive insights from mobile money transactions.

- Real-time analytics is used for fraud detection, enabling immediate identification of suspicious activities such as high-value transactions, unusual transaction times, and abnormal user behavior patterns.
- Batch analytics supports trend analysis and reporting, including transaction volumes, regional activity, and user spending patterns.

Data is queried using Trino and visualized through Apache Superset dashboards.

Fraud detection is based on simple rule-based logic, such as:

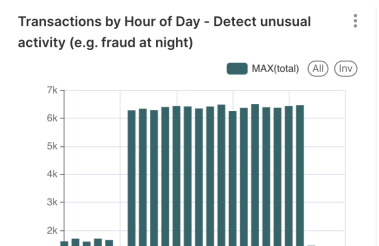
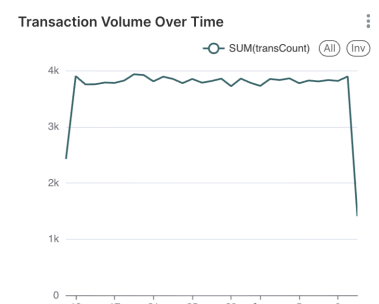
- Transactions exceeding predefined thresholds
- International or cross-region anomalies
- Unusual activity during off-peak hours

These analytics enable proactive monitoring, faster decision-making, and improved financial risk management.

Example queries:

i) Transaction volume over time

```
SELECT date_trunc('day', CAST(event_date AS TIMESTAMP)) AS event_date,
       sum("transCount") AS "SUM(transCount)"
FROM (SELECT date(timestamp) AS event_date, COUNT(*) as transCount
      FROM hive.finance.mobile_transactions
      GROUP BY date(timestamp) ORDER BY event_date) AS virtual_table
GROUP BY date_trunc('day', CAST(event_date AS TIMESTAMP))
ORDER BY "SUM(transCount)" DESC LIMIT 1000;
```

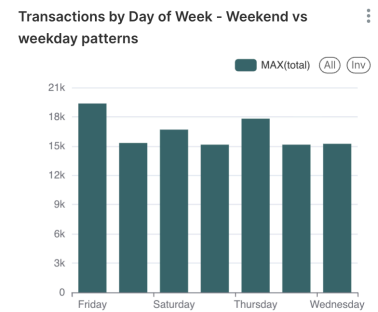


ii) Transactions by Hour of Day - Detect unusual activity (e.g. fraud at night)

```
SELECT hour AS hour, max(total) AS "MAX(total)" FROM
(SELECT hour(timestamp) AS hour, COUNT(*) AS total
FROM hive.finance.mobile_transactions
GROUP BY hour(timestamp) ORDER BY hour) AS virtual_table
GROUP BY hour ORDER BY "MAX(total)" DESC LIMIT 1000;
```

iii) Transactions by Day of Week - Weekend vs weekday patterns

```
SELECT day AS day, max(total) AS "MAX(total)" FROM
(SELECT format_datetime(timestamp, 'EEEE') AS day, COUNT(*) AS total
FROM hive.finance.mobile_transactions
GROUP BY format_datetime(timestamp, 'EEEE')
ORDER BY min(day_of_week(timestamp))) AS virtual_table GROUP BY day
ORDER BY "MAX(total)" DESC LIMIT 1000;
```



5. Governance, Security & Feasibility

The data lake implements strong governance and security controls to protect sensitive financial data and ensure compliance with the Kenya Data Protection Act 2019.

Identity and access management is handled using Keycloak, which provides centralized authentication, role-based access control (RBAC), and secure user management across all components (Kafka, Spark, Trino, and Superset). This ensures that only authorized users can access specific datasets and analytics tools.

Data is further secured through encryption at rest and in transit, while audit logs track user activity for accountability and compliance. Governance is supported through metadata management and data lineage tracking, ensuring data quality and transparency. To enhance privacy, personally identifiable information (PII) is minimized or anonymized. Overall, the solution is secure, compliant, and feasible, providing controlled access and protection of mobile financial data.

References:

MinIO. (2024). MinIO Object Storage Documentation. Retrieved from <https://min.io/docs>